

Monet v1.0: imitating a frontier Canadian Fish agent, correcting the imitation, and passing a pre-registered acceptance test

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Abstract

Monet is an agent for Canadian Fish, the six-player team card game, under the **us54** rules. It began on 2026-09-01 as FishAI’s Bass v2.0, which won 27.08% of its games against SESTINA v1.0, the frontier agent of the FishLab lineage, and it climbed through fifty-five rungs, each registered before it was measured and each measured against SESTINA in FishLab’s own engine. Version 1.0 was defined in advance by six conditions rather than by a date. The vector that meets them asks with a small network that imitates SESTINA’s choice, overruled by a second network trained on paired play-outs of what wins; its belief is hand-built constraint tracking with a fitted card-by-seat table, its declarations are hand-written, and it searches nothing at play time. On the acceptance read it won 58.38% of 14,400 games against SESTINA on twelve fresh seeds, with no seed below 56.58%; it beat each of five earlier FishLab agents; it declared correctly on at least 99.49% of attempts in every cell; and an independently built adapter reproduced all 14,400 of its games against SESTINA byte for byte. We report the rungs that failed as fully as those that shipped: no play-time search beat the agent it searched over against SESTINA, and a search over every legal ask on Monet’s own belief was worth -0.14 sets at a decision against a hindsight ceiling of $+2.84$. We also report that the read’s locked control reader printed “not met”, and how the registration’s own rules decide it.

1 Introduction

Canadian Fish is a game of asking. Six players in two teams of three hold a shuffled deck; on a turn a player asks one opponent for one card, and keeps the turn while the asks hit. A team scores a half-suit by declaring, from memory and inference, which teammate holds each of its six cards, and a wrong declaration hands the set to the other side. The information is public but scattered across a long log, the hands change with every hit, and the value of an ask lies less in the card than in what it reveals and what it sets up.

This paper is the final report on Monet, the agent line FishAI built to answer one question the project’s owner set on 2026-09-03 and sharpened on 2026-09-10: can a FishAI agent be *consistently* better than SESTINA v1.0? SESTINA is the frontier agent of FishLab [4], an independent engine and bot family whose C++ engine plays a guest bot over a JSON protocol. Every number in this paper is a game played on that engine, against that opponent, through a bridge described in §2.3; FishAI’s own engine is used only as a cheap screen.

How the work was done. Monet was built as a ladder. Each rung wrote its mechanism, the bar that would ship it, a probability for each prediction, and its cost into the project’s research record, MONET.md [1], before the first game was played, and each rung’s outcome was recorded whether or

not it shipped. Twelve vectors shipped. Most rungs did not, and the record of why is the bulk of what the project learned (§6).

What the paper reports.

1. *The agent* (§4): a belief built from constraints and a fitted table, a hand-written ranker whose numbers become features, a clone of SESTINA’s asks, and a learned advantage that overrules the clone when it expects more than 0.05 sets from another ask.
2. *The ladder* (§5): from 27.08% to 58.42% against SESTINA in four phases — belief, ranker terms, imitation, and the learned correction.
3. *The negatives* (§6): six forms of play-time search, the belief after v0.4c, and the selectors over the clone’s shortlist, all measured and all at or below zero.
4. *The game* (§7): what the records say about Fish between strong agents, beginning with the exchange rate of one set a game for 12.8 points of win rate.
5. *The acceptance read* (§8): the six conditions at v0.54’s vector, on 100,800 games, with the one condition whose locked reader disagreed set out in full.
6. *The comparison* (§9) with Bass v2.0, SESTINA v1.0 and Kraken v1.0, and the limits of every claim here (§10).

Writing the paper found errors in our own record; §11 lists them and where MONET.md now corrects them.

2 The game, the opponent and the bridge

2.1 Canadian Fish under us54

Canadian Fish (Literature) is a six-player game of two teams of three, seated alternately. **us54** [5] is the “US student” dialect FishAI plays: all 54 cards (the standard deck with the eights and two distinguishable jokers) are dealt, nine to a player, and form nine half-suits of six — low and high in each suit, plus a ninth of the four eights and the two jokers. On a turn a player asks one opponent for one specific card, and may ask only for a card of a half-suit in which they hold at least one card and do not hold the card asked for. A hit transfers the card face up and the asker asks again; a miss passes the turn to the player who was asked. At any time, in or out of turn, a player may *declare* a half-suit by naming the teammate who holds each of its six cards. A fully correct declaration scores the set for the declarer’s team; *any* error — an opponent holding one of the six, or a card assigned to the wrong teammate — scores it for the opponents. The game ends the moment a team has five sets, so ties cannot occur. Card counts and the full log of asks, answers and declares are public; the hands are not.

Three consequences of these rules shape everything below. *Every ask is evidence*: the asker has revealed a card of the half-suit (the *licence*), and a miss proves the target lacks the card. *A wrong declaration is a two-set swing*: the set a correct declaration would have scored goes to the other side instead. *A set the team fully holds is safe*: the opponents hold none of it and can never ask into it, so a certain set loses nothing by waiting.

2.2 The opponent: SESTINA v1.0 and the panel

SESTINA v1.0 is the frontier agent of the FishLab project [4], which publishes a C++ engine, a family of bots and an evaluation laboratory for the same game. Every number in this paper is Monet playing SESTINA’s frozen release specification, verified byte-identical to the release asset (MONET.md §6.1):

```
v07:r12=25,rtie=1,pool=-1,oppfloor=-1,force=1000000,askfloor=-1,stall=12,  
s1=1,det=12,cand=4,kappa=2.5,rbelief=indep,depth=12,maxq=26
```

From FishLab’s documentation and engine source (read for facts, restated here in our own words): SESTINA chooses an ask by a linear score over twenty features of each legal ask, with weights fitted by cross-entropy search, and a one-step lookahead over that score. The keys **s1**, **det**, **cand**, **depth**, **kappa** and **maxq** configure a determinized search that runs only in the endgame, when at most 26 cards are unresolved: it samples 12 deals consistent with the public record, plays each of the top 4 candidate asks out 12 moves under a blueprint policy (whose belief **rbelief** names), and departs from the blueprint’s own pick only when a candidate beats it by 2.5 paired standard errors. It declares a set when its team-ownership probability for the plan is at least 0.849 and a value check agrees; on our records it declared with cards it could not place 0.53 times a game, right 84.1% of the time (MONET.md §3.8aj).

The *panel* is FishLab’s earlier releases v0.2 to v0.6, which neither adapt nor, from v0.5 on, search. They are in the acceptance test for one reason, stated when it was written: a bot that beats SESTINA and loses to v0.5 “has been fitted, not built” (MONET.md §3.9).

2.3 The bridge

Monet is measured *abroad*: FishAI’s decision function is installed as a guest bot inside FishLab’s engine and speaks its documented JSON-lines protocol through an adapter that builds FishAI’s public view from the host’s events. FishLab’s repository carries no licence file, so the engine, the recording fork, the adapters and the game records live outside FishAI’s public repository and nothing of FishLab’s is reproduced here; its game output is data, and its measured findings are cited as its own. Two dialect differences are translated at the adapter, never in the policy. **us54** has no pass, and the host offers one to a turn-holder with no cards (PASSFIX: take it). **us54** compels a declaration when every opponent is out of cards, and the host offers an optional poll instead (MUSTFIX, adopted on 2026-09-04: an uncertain compelled claim is answered **none** and the host’s own forced endgame resolves it; worth +1.28 points paired on v0.9’s vector, ahead on 12 of 12 seeds, MONET.md §3.8f). The host plays all nine half-suits where **us54** stops at five; the winner is the same.

Cells, reads and floors. A *cell* is one seed: 200 duplicate deals, each played in all 6 seat rotations, 1,200 games. A *read* is twelve cells on fresh seeds, 14,400 games, quoted as the mean over seeds with the seed-to-seed SD and, for a comparison, the paired standard error of the per-seed differences. The engine’s own power line puts the paired floor at ± 6.93 points for one seed and ± 2.00 for twelve. v0.54’s read played 800 deals a seed (4,800 games a cell), because at 200 a true +0.7-point gain would clear the bar only 29% of the time (MONET.md §3.8az). Seeds are drawn by the engine’s own hash from a label written before the cell and are never reused; 259 were on file when the acceptance read drew its twelve.

Pins and controls. Before any number is read, every recorded ask of every cell is replayed at home through the registry’s own policy and must agree at every ask; one cell replayed by the previous

version must differ, which proves the pin can see the vector. A new arm or tree first reproduces a known cell game for game (*step 0*). The mandatory controls (MONET.md §6.2) are op coverage against a written per-op expectation, a byte-exact null arm, the identity pin, paired deals, named fault counters read at zero, a calibration harness on every cell, a completion check, and a home regression before any ship. The mirror cell is deliberately not one of them: a policy against itself on duplicate deals reads 50% by construction, and it once read a perfect 50.0000% across a 3.44-point defect [3].

3 The protocol

Pre-registration. Every rung of the ladder is written into MONET.md before its first cell: the mechanism, the bar that ships it, a probability for each prediction, and the cost. Predictions are scored after the read, hits and misses alike, and a finding that was not registered is marked as such. Up to v0.13 a rung was fitted on three fresh seeds, which may only *choose* an arm, and priced on twelve more. The project’s owner makes every ship decision; the record says which were theirs.

The ship rule. On 2026-09-06 the owner fixed the standing objective in one sentence: “*just pick whatever will improve the winning percentage of Monet*” (MONET.md §3.8n). A term ships when a pre-registered read on twelve fresh seeds, pinned in-engine at 100%, puts its paired mean at least two standard errors above zero and the candidate ahead on a majority of seeds (from v0.53, strictly ahead on at least 7 of 12). The ± 2.00 floor stays the definition of a *rung*, and a term that ships under it is marked as such.

Home reads. FishAI’s own engine plays duplicate pairs at home as a cheap screen. Two lessons govern them. A home set difference counts the sets declared by the five-set finish, so a knob that only declares sooner reads a gain without winning more games; timing knobs are read by the pairs’ win rate. And a home opponent built from Monet’s own policy is not SESTINA: one search arm read +3.4 standard errors at home and -2.1 abroad (§6).

The exchange rate. Over the 86,400 games of the v0.33 panel read, the win rate is almost exactly linear in the sets-a-game differential:

$$\text{win rate} = 50.449 + 12.785 \times (\text{sets a game, ours} - \text{theirs}), \quad R^2 = 0.9942.$$

One extra set a game is 12.8 points of win rate (MONET.md §3.9).

Version 1.0 is a test, not a date. MONET.md §3.9 defines Monet v1.0 as the vector that satisfies six conditions, all at once, on one pre-registered read:

- C1** a win rate of at least 52.0% against SESTINA v1.0, pooled over twelve seeds of 200 deals, paired floor ± 2.00 ;
- C2** every one of the twelve seeds at least 50.0%, every seed reported and the SD published;
- C3** the panel is *monotone*: each of v0.2 to v0.6 beaten (a twelve-seed mean of at least 52.00) and the win rates ordered $v0.2 \geq v0.3 \geq v0.4 \geq v0.5 \geq v0.6 \geq \text{SESTINA}$, ties inside the floor allowed;
- C4** declaration accuracy at least 98.0% on every cell, and zero fault counters across the run;

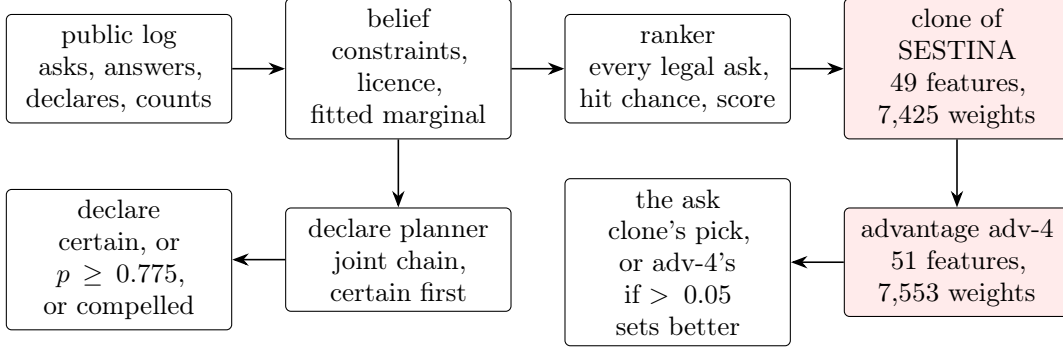


Figure 1: How Monet v1.0 decides. Every stage reads only public information. The two shaded stages are learned; everything else is hand-written. Nothing is simulated at play time.

C5 every control of §6.2 passes, including op coverage, and the mirror cell is not among them;

C6 the result is reproduced by a second, independently built adapter.

Conditions 1 and 2 were amended on 2026-09-10 on the owner’s word — “*I want Monet to become the new frontier bot, which means being consistently superior to SESTINA*” — from “at least 50.0%” and a reporting rule. The record priced the amendment when it was made: at the seed-to-seed SD of 1.63 points, twelve seeds all at or above 50% needs a pooled mean near 53–54%, so condition 2, not condition 1, is the binding one.

4 The agent

Monet v1.0 is the registry entry of v0.54 under a new name: Bass v2.0’s *Punter* style at the **hard** skill preset, with ten knobs moved (Table 1). It searches nothing at play time. A decision costs 0.178 ms on average and a whole game 119 ms (MONET.md §3.8az, measured on an AMD Ryzen 9 9900X), against budgets of 1.4 ms and 0.9 s.

4.1 Belief

Monet keeps, for every unseen card, the set of seats that can still hold it: a card leaves a seat’s candidates when the seat is asked for it and misses, when it is seen elsewhere, or when the seat’s hand is full of known cards. The licence adds weight to the asker’s holding of the half-suit, at `licenceLambda`. Under `pModel: 'marginal'` (v0.4a) the constraints become a card-by-seat probability table fitted by iterative proportional fitting, so each seat’s row sums to its hand size and each card’s column to one. Under `pAssignment: 'joint'` (v0.4b) the declare planner places a set’s open cards by a chain over that table, most certain first, and a plan’s probability is the product of the conditionals rather than of the marginals. The belief is the work of v0.3 to v0.4c; every later attempt to improve it measured at or below zero (§6).

4.2 The ranker

The hand-written ranker lists every legal ask with its hit probability p and a score,

$$70p + 18 \cdot \text{progress} + 12 \cdot \text{narrowing} + 20 \cdot \text{certain} + \text{gamble},$$

where *progress* is the fraction of the half-suit already certainly on the asker’s team, *narrowing* is how much a miss would narrow the card’s candidate seats ($1/(n-1)$ when the target is one of n

candidates), *certain* marks a sure hit, and Punter’s *gamble* bonus of 25 is paid when a hit would complete the set. Before v0.30 the ranker’s argmax was the ask; from v0.30 on, the ranker only lists the asks and its numbers become features of the learned stages.

4.3 The clone of SESTINA

The ask is chosen by `sestina-clone-3`, a model of SESTINA’s own choices (MONET.md §3.8ac, §3.8af). Each legal ask is described by 49 public features: 33 from the ranker and the log (the hit probability and its certainty, the ranker’s relative score and rank, the half-suit’s picture by side and by target, the target’s hand, who opened the half-suit, whether the card was asked before, the seat’s run and last target, and the state of the game), and 16 that hand the model the stack’s own belief and the target’s dealings with the half-suit (the slot prior, an independent per-card probability, the target’s unknown slots, and the asks, hits, misses and transfers at the target in the half-suit and how long ago). A multilayer perceptron of width $64 \cdot 64$ (7,425 weights) scores each ask as a conditional logit over the legal set, and the choice is the argmax. It was fitted on 1,160,665 of SESTINA’s ask decisions from the bridge records (a 5% sample), and it agrees with SESTINA’s choice on 58.51% of held-out decisions, against 56.56% for the same network on the first 33 features and 43.73% for the stack’s own choice. The fit says what SESTINA does: it keeps asking into the half-suit it or its partner asked into before, at the target it asked last, from the smaller hands, and it takes a certain hit for its probability and no more (MONET.md §3.8ac).

4.4 The learned advantage

The clone arrives at SESTINA and stops there, so the last stage is trained on what *wins* rather than on what SESTINA chose (MONET.md §3.8aw–§3.8az). The data are paired play-outs from the true deal: at a sampled ask decision of a self-play game, the clone’s ask and an alternative are each played to the end under one random key, and the label is the difference in the asking team’s final set differential. Each pair sets the ask actually played against one alternative: the clone’s next three, the ranker’s top two, three at random, and — in the later rounds — the previous model’s own best, so that the model is corrected exactly where it over-rates. `adv-4` saw 3,025,321 such pairs at 554,197 decisions of 32,000 self-play games, in rounds at the own choices of `adv-1`, `adv-2` and `adv-3`. It is a $64 \cdot 64$ network over 51 features — the clone’s 49, the clone’s own score relative to its best, and whether the ask is the clone’s top — with 7,553 weights, fitted with the settings registered at `adv-1` (Adam, batch 256, learning rate 10^{-3} , L_2 10^{-5} , 12 epochs, the best held-out epoch kept, every game $\equiv 0 \pmod{5}$ held out).

At play, `adv-4` scores every legal ask, and the clone’s choice is replaced only by `adv-4`’s best, and only when `adv-4` rates it more than 0.05 sets above the clone’s own. The margin was chosen on one half of 8,000 fresh games and scored on the other: there it gained +0.0219 sets a decision (z 3.70), and it departs from the clone’s choice at 38.14% of decisions. Replayed through v0.53’s policy, the first cell of the v0.54 read differs at 35.9% of Monet’s asks.

4.5 Declares, passes and the rest

The declare has never been on the learned path. A seat that can place all six cards declares the set. Below certainty Monet guesses only *who* on its own team holds a card, never *whether* the team holds it: the speculative branch runs only when every open card’s candidate seats are teammates, and then only when the joint plan’s probability is at least 0.775 (0.5 when the game has stalled), with at most two uncertain cards, and at 0.975 for a set in which the seat holds no card. When the rules compel a declaration it plays the planner’s best plan, and abroad an uncertain compelled claim

Table 1: Monet v1.0 = v0.54. Every knob, the rung that added it, and the read that shipped it (win rate against SESTINA; paired difference over the version before, on the same fresh seeds).

knob	what it does	added	shipped on
Punter at hard	Bass v2.0’s style: ranker weights, declare bars 0.775 / 0.5 / 0.975	v0.1	27.08% (the baseline)
minHitP 10^{-9}	sets aside asks that cannot hit	v0.2	correctness; “must not move”
licenceLambda 0.3	licence weight on the ask’s p (inert since v0.30)	v0.3, v0.4c	+1.19 at 0.3 vs 0.6, 24 seeds
pModel ‘marginal’	the fitted card-by-seat table	v0.4a	31.94% (six seeds)
pAssignment ‘joint’	the declare planner’s chain	v0.4b	+0.81; declares 97.86 $\rightarrow$ 98.32%
contest 0.6	credit for contested asks likely to miss (inert since v0.30)	v0.9	+4.04, 12 of 12
closing 0.5, closingFour 2	credit for asks that close a set (inert since v0.30)	v0.20c	+0.94, 8 of 12
askModel	the SESTINA clone chooses the ask	v0.30, v0.33	+7.24; then +3.64
askAdvantageModel ‘adv-4’	the learned override	v0.53, v0.54	+1.46 (adv-2); then +6.25
askAdvantageMargin 0.05	how much better the override must be	v0.54	(as above)

is left to the host (MUSTFIX). The 0.775 bar was priced at v0.50 against the true deal and sits where realised correctness crosses break-even (MONET.md §3.8av). Four knobs that shipped earlier — `licenceLambda`, `contest`, `closing` and `closingFour` — act on the ranker’s own choice and are therefore inert since v0.30, when the clone took that choice over. They stay in the vector because removing them would be a new vector.

5 The ladder

Fifty-five numbered rungs separate v0.1 from the acceptance read (there is no v0.48, and v0.3, v0.5 and v0.20 each carry more than one entry). Twelve vectors shipped; everything else was measured and recorded whether or not it shipped. Table 2 lists every shipped vector and the read that shipped it, and Figure 2 draws the line.

Four phases. The line climbed in four phases, and each was bought by a different kind of mechanism.

1. *Belief*, v0.3–v0.4c (27.08 $\rightarrow$ 35.12%). Licence conditioning, the fitted marginal and the joint declare chain. The record’s reading of the licence term is that it buys tempo, not accuracy: it asks into the sets the team is invested in, and its ask accuracy *fell* (53.84 $\rightarrow$ 53.04%) while the win rate rose (MONET.md §3.4c).
2. *Ranker terms*, v0.9 and v0.20c ($\rightarrow$ 41.25% on the translated bridge). The contest credit pays for asks that will probably miss into a set the opponents dominate and nobody can yet place. It moved 15.2% of Monet’s asks, cut the asks into sets its own side already held from 8.9% to

Table 2: Every shipped vector. *Read*: win rate against SESTINA v1.0 on the rung’s own fresh seeds. *Paired*: the mean per-seed difference over the named base on the same seeds, with its SE and the seeds on which the candidate was ahead. Levels before 2026-09-04 were read before MUSTFIX and sit about 1.3 points low; the paired steps stand (MONET.md §3.8f).

version	what it added	read	paired over the base	seeds $\times$ deals	MONET.md
v0.1	Bass v2.0, unchanged	27.08%	—	6×200	§3.1
v0.2	dead asks set aside	27.21%	+0.13 (“must not move”)	6×200	§3.2
v0.3	licence at $\lambda = 0.6$	30.96%	+3.75, 6 of 6	6×200	§3.3a
v0.4a	the fitted marginal	31.94%	+0.99, 3 of 6 (unresolved)	6×200	§3.4a
v0.4b	the joint declare chain	32.75%	+0.81 (inside the floor)	6×200	§3.4b
v0.4c	licence back at $\lambda = 0.3$	35.12%	+1.19 (SE 0.30), 19 of 24	24×200	§3.4c
v0.9	the contest credit	38.92%	+4.04 (SE 0.47), 12 of 12	12×200	§3.8d
v0.20c	the closing credits	41.25%	+0.94 (SE 0.43), 8 of 12	12×200	§3.8s
v0.30	the SESTINA clone	48.01%	+7.24 (SE 0.63), 12 of 12	12×200	§3.8ac
v0.33	the clone’s second feature set	51.04%	+3.64 (SE 0.71), 11 of 12	12×200	§3.8af
v0.53	the learned advantage (adv-2)	52.05%	+1.46 (SE 0.52), 8 of 12	12×200	§3.8ay
v0.54	adv-4 at margin 0.05	58.42%	+6.25 (SE 0.24), 12 of 12	12×800	§3.8az

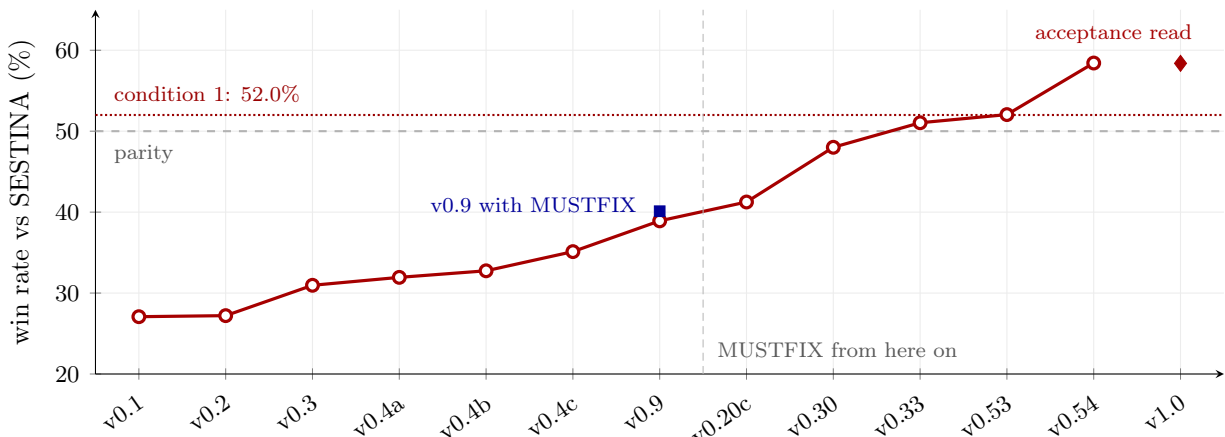


Figure 2: Monet’s win rate against SESTINA v1.0 at every shipped vector, each on its own fresh seeds. The square is v0.9’s vector re-read on the translated bridge, where every later level is read. The diamond is v1.0 — v0.54’s vector, unchanged — on the acceptance read’s twelve fresh seeds (§8). The two steepest steps are imitation (v0.30) and the learned correction (v0.54).

5.9%, kept the turn longer (41.7 asks a game from 39.2) and took the evenly split sets from 42.0% to 46.1% (MONET.md §3.8d). The idea is FishLab’s own attribution of SESTINA, whose largest scoring component rewards exactly such asks [4]; the term is our implementation of it.

3. *Imitation, v0.30–v0.33* ($\rightarrow 51.04\%$). The clone was the largest single step of the ladder, +7.24 points (12 of 12 seeds; +7.13 at home), and the second feature set added +3.64. Pooled over twenty-four seeds v0.33 read 50.46%: parity.
4. *The learned correction, v0.53–v0.54* ($\rightarrow 58.42\%$). The first advantage model, adv-2 at margin 0.2, changed the clone’s choice at a few asks in a hundred and gained +1.46. Four times the data, a third round of labels at the model’s own choices and a margin of 0.05 made adv-4, which changes more than a third of the asks and gained +6.25 on twelve fresh seeds at 800 deals (26.5 SE, 12 of 12, the lowest seed 57.31%), the second largest step. At home it won

55.76% of 25,600 games head to head with v0.53.

In sets: the paired set differential moved +0.430 sets a game at v0.54, which the exchange rate prices at +5.50 points against the +6.25 read. Monet’s own asks hit more often (50.69 → 52.63%) and it won more sets (4.57 → 4.79 a game); the record calls these descriptions, not a mechanism.

6 What was measured and did not ship

Most rungs shipped nothing. Table 3 groups them by the idea they tested; the sentences after it say what each group taught.

Search. Six forms of play-time search were priced, and none beat the policy it searched over against SESTINA (MONET.md §3.8a, §3.8aa, §3.8ac, §3.8ai, §3.8au). The one that looked best at home, where its rollouts modelled the opponents exactly, lost abroad, where they did not. Giving the rollouts a model of SESTINA did not fix it, and neither did a better leaf: the learned leaf predicted the end better and played worse, because it walked away from certain hits. v0.50 then bounded the question. With the true deal, the best legal ask is worth +2.841 sets over the pick at a decision; with Monet’s own belief, a perfect search over every legal ask finds −0.141. *Hindsight is worth more than the whole ceiling*, and nothing a belief-limited search can see is left on the table at the ask.

Belief. After v0.4c the belief was studied mainly on the records, where the true deal is known, and every route into it was priced at zero or below. The side assignment is calibrated; a seat-known majority credit buys sure misses, not chases; and the calibrated licence likelihood, real on the records, doubled the wrong declarations through the declare chain abroad. A pooled calibration score could not see that selected population (MONET.md §3.8j–§3.8l).

Imitation caps at the imitated. Every clone was trained to predict SESTINA’s choice, so it arrives at SESTINA and stops there; the record reached that conclusion at v0.33’s parity and tested its obvious remedy — a selector over the clone’s shortlist — three ways (by the belief, by search, by a value fitted on winning), and all three lost (MONET.md §3.8as). What got past parity was not a selector over a shortlist but a correction over *every* legal ask, fitted on paired outcomes from the true deal and gated by a margin.

The winner’s curse. The first advantage model was right on average and wrong where it acted: over its 53,263 departures from the clone it predicted +0.092 sets and the true deal gave +0.005. Labels at the model’s own choices (a second and a third round) and four times the data are what made the correction act on a third of the asks rather than one in seventy; the record does not separate their contributions.

7 What the record says about the game

These are findings about us⁵⁴ Fish between strong agents, each measured on the records or at the bridge; the section cites where.

- **One set a game is worth 12.8 points of win rate (§3)**, so a mechanism is worth what it moves in sets.

Table 3: The rungs that shipped nothing, grouped by idea. *Abroad* is against SESTINA on twelve fresh seeds (paired, points, ± 1 SE unless marked); *home* is duplicate pairs in FishAI’s own engine; a records study reads the bridge records and plays no cell.

rung	mechanism	what it read
<i>Search at play time</i>		
v0.7	determinized search, lock-only leaf, ~ 100 ms	+0.14 (SE 0.67) abroad: a no-op
v0.28	an ask the search at any compute; 32 deals (t32)	+0.248 a pair at home (+3.4 SE), then -1.61 (SE 0.77) abroad
v0.29	a learned leaf for the search	eligible at home by a fifth of an SE; never read abroad
v0.30 M4	the clone as the rollouts’ opponents	-1.33 (SE 0.74) abroad
v0.36	the clone at every seat of the rollouts	-1.26 points (-2.2 SE) at home
v0.44	the search over the clone’s shortlist	stopped at its gate: the shortlist held a hitting ask 84.0% of the time and the search still chose worse
v0.49	the learned leaf inside the search	the marker against the true deal -0.034 (SE 0.021); overrode a certain hit 27% of the time at -0.184 sets each
v0.50	a <i>perfect</i> full-legal-set search on Monet’s own belief	-0.141 (SE 0.102) sets, against a hindsight ceiling of +2.841
<i>Belief after v0.4c</i>		
v0.5	opponent reading (a choice prior)	+0.47 (SE 0.59) abroad
v0.14–v0.15	the assignment; the deduction fix	records studies: both axes closed
v0.16	the licence likelihood, calibrated	-0.74 (SE 0.46); wrong declares doubled through the chain
v0.34–v0.35	a wider clone; a holder model of the deal	stopped at their fit gates (+0.02 and -0.16 of top-1)
v0.42–v0.43	the greedy ask by the belief; its repair	the ranker’s top is the greedy player, which lost to the clone by 7.24
<i>Ranker terms</i>		
v0.10	a charge for hits the opponents can take back	+1.07 (SE 0.54); re-read -0.15 (SE 0.67)
v0.12–v0.13	the closing credit by belief; a chase appetite	-0.47 to -2.17 ; -1.08 to -7.56 on the fit
v0.20b, v0.22–v0.27	the closing family, every rung and dose, one and two at a time	stopped at home: the shipped point is a local optimum on every axis
<i>Declaring</i>		
v0.8	declare as soon as sampled deals agree	+0.08 (SE 0.08) abroad
v0.11	a risk bar on the speculative declare	priced on the records at a quarter of a point; not built
v0.32	a clone of SESTINA’s declares	per completed set, SESTINA declares as Monet does; not built
v0.38	bolder speculative claims (the holder model at the claim)	+0.165 a pair at home (tempo), $+0.28 \pm 0.50$ (2 SE) abroad
<i>Communication and learned values</i>		
v0.6	asks chosen to reveal; the handoff	flat or behind at home
v0.31	the clone refitted wider and longer	+0.027 a pair (SE 0.150) at home
v0.46	a value fitted on winning, over the clone’s top k	-0.592 and -0.209 a pair at home (8.97 and 3.18 SE behind)
v0.51	the first advantage model (adv-1)	its marker missed near zero (z 1.01): the winner’s curse

- **The position decides more than the ask.** Over v0.33’s self-play, state features alone explain $R^2 = 0.295$ of the final set differential and the playable ask columns alone 0.018; the bare scoreboard predicts the end (R^2 0.248) as well as a 24-step rollout (0.249) (MONET.md §3.8as, §3.8au).
- **A hit’s chance is not an ask’s value.** The closing credit took asks that hit seven points less often and won; SESTINA leaves hits on the table and wins; hitting more is not evidence of playing better (MONET.md §3.8c, §3.8h, §3.8as).
- **A certain hit is worth keeping.** The ranker’s certainty margin is worth about two points of win rate per point of dose (MONET.md §3.8i), and the search lost where it walked away from one.
- **Fish where the fish are.** Asks into sets the opponents dominate and nobody can yet place win races even when they miss (the contest credit, §5).
- **A split set is decided at four of six.** The first side to hold four of a 3–3 set converts it 56% of the time between equal players; before the clone, SESTINA chased first at 59.4% of its lead decisions where Monet’s picker would have at 39.9% (MONET.md §3.8p, §3.8q). Under the clone the gap closed: leads converted 61.1% and 60.9% (§3.8an).
- **Declaring early buys nothing; a wrong declaration costs two.** A set the other side got was never a lock, because the licence rule means a lock cannot be broken; a certain set cashed later pays the same (MONET.md §3.8f). Monet’s 0.775 speculative bar sits where realised correctness crosses break-even (§3.8av), and every experiment that made Monet declare sooner or more boldly failed to win more games.
- **A hindsight oracle is not a target.** The full-information best ask is worth +2.84 sets at a decision; none of it was reachable by a search on Monet’s own belief. The reachable gain was found by learning from outcomes, not by searching the belief.

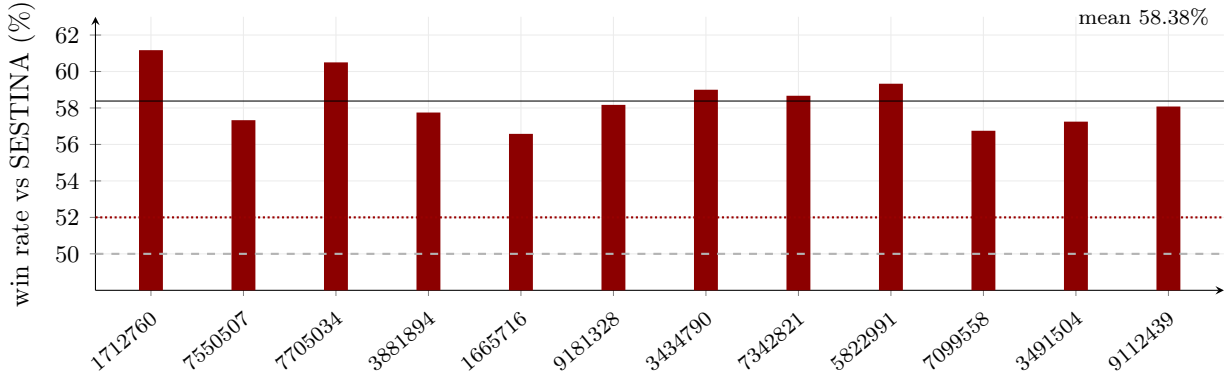
8 The acceptance read

The read was registered in MONET.md §3.8ba before any cell was played, on the owner’s answer to the proposal that closed the previous decision (“*we’ll go with your suggested next step*”). It measures the shipped vector, v0.54, under the six conditions of §3 exactly as they were written, and ships nothing. The three scripts that score it were adapted from the earlier reads’ and run on those reads’ cells, last written before the first cell of this read finished, and fixed by md5 before any cell was read; the runners’ consoles print completion only.

What was played. Twelve fresh seeds, drawn under a label written before the read against every seed on file. On each: the arm against SESTINA v1.0 and against each of v0.2 to v0.6, and the independently built arm against SESTINA, 200 deals $\times$ 6 rotations = 1,200 games a cell. That is 84 cells and 100,800 games, played on FishLab’s engine through the bridge, two containers at a time, in about one and a half hours of wall clock. The arm plays the registry’s v0.54 entry from an export of the shipped tree, with no model file and no override, so it plays exactly what the web build plays. Before anything else, a step-0 cell pinned the new arm to the cell that shipped v0.54 (seed 4549308, 800 deals): every engine line but the header and the elapsed time was identical. After the games, every recorded ask of all 84 cells was replayed in-engine through the registry’s v0.54 entry, and every cell agreed at every ask (72 of 72 for the arm, 12 of 12 for the twin); the same replay by v0.53 differs at 35.5% of the asks of the cell it was run on, so the check can see the vector.

Table 4: The six conditions of MONET.md §3.9 at v0.54’s vector, on twelve fresh seeds (§3.8ba).

	condition	read	verdict
C1	$\geq 52.0\%$ against SESTINA, twelve seeds, floor ± 2.00	58.38% over 14,400 games (56.38 to 60.38 with the floor)	met
C2	every seed $\geq 50.0\%$, SD published	lowest 56.58%, highest 61.17%, SD 1.43	met
C3	v0.2 to v0.6 beaten, and the order monotone	84.43/81.20/56.91/57.92/58.24 against SESTINA’s 58.38; three ties inside the floor	met
C4	declare accuracy $\geq 98.0\%$ on every cell; zero fault counters	99.494 to 99.857% over 72 cells; the fourteen counters zero in 2,587 process files	met
C5	every control of §6.2, the mirror cell excluded	the locked reader: <i>not met</i> on two rows; both resolved as registered (below)	met as registered
C6	reproduced by an independently built arm	58.3819% against 58.3819%, every seed equal; 14,400 of 14,400 games byte-identical	met

**Figure 3:** The twelve seeds of the acceptance read against SESTINA v1.0, 1,200 games each, in the order they were drawn. The solid line is the mean, 58.38%; the dotted line is condition 1’s bar for the mean (52%) and the dashed line condition 2’s bar for every seed (50%). The lowest seed is 56.58%.

Conditions 1 and 2. Against SESTINA the arm won 58.38% of 14,400 games, with every seed between 56.58 and 61.17% (Figure 3); the SE across seeds is 0.41 points. The shipping read of v0.54 had read 58.42% on twelve other seeds at four times the deals. The stricter reading that the record reports beside condition 1, that the floor’s lower end clears 52.0%, holds too. Pooled, Monet declared 4.64 sets a game at 99.76% and SESTINA 4.29 at 96.92%; Monet won 4.80 sets a game to SESTINA’s 4.20, a differential of +0.595 that the exchange rate prices at 58.06%.

Condition 3. Every older bot is beaten (Table 5). The order is monotone only through ties: v0.4, v0.5, v0.6 and SESTINA sit within 1.47 points of one another, and Monet’s win rate rises slightly from v0.4 to SESTINA, the reverse of the order the condition names; every adjacent pair is inside the ± 2.00 floor, which the condition counts as a tie. The registration named this condition as the one most at risk: if the learned correction had gained against SESTINA alone, SESTINA’s column would have sat more than the floor above v0.6’s and the order would have broken. It did not. Against v0.4 to v0.6 the win rate rose from about 54% at v0.33 to between 56.9 and 58.2%; the seeds differ, so the comparison is not paired.

Table 5: The panel. Twelve seeds of 1,200 games per opponent. The last column is the v0.33 panel read (§3.8ar), on twelve other seeds.

opponent	mean	SD	lowest	highest	v0.33
v0.2	84.43%	0.80	82.92	85.58	85.42%
v0.3	81.20%	0.59	80.08	82.17	79.28%
v0.4	56.91%	1.02	55.67	58.67	54.67%
v0.5	57.92%	1.27	56.00	59.75	54.01%
v0.6	58.24%	1.26	55.25	60.42	53.99%
SESTINA	58.38%	1.43	56.58	61.17	49.87%

Condition 4. The worst declare cell of the 72 is 99.494%. The adapter’s fourteen named fault counters are zero in every one of the 2,587 per-process files that were written (5 of 2,592 were lost to the collector: one torn, four never written); no FATAL line was logged and no cell hit the host’s move limit.

Condition 5, and the reader that said no. The locked reader for §6.2’s controls printed “*CONDITION 5 NOT MET*”. Every row passed but two, and the registration had already said how each of those is decided.

- *Twelve op counts fell outside the written expectation.* The registration wrote a band for each op before the read and said that a miss passes only when it is located to a cause that is neither a defect of play nor of the instrument; the reader cannot locate, so it fails the row. Ten misses are `mustfixDeclines` against v0.2 and v0.3, whose bands came from v0.33’s cells. The adapter counts a decline only where FishAI’s rules would compel a claim and the claim is not certain, and the engine’s own records show where that happens: v0.54’s games against those two bots reach FishAI’s dead-position rule before the clinch about 2.6 times as often as v0.33’s did (0.346 positions a game against 0.134, and 0.118 against 0.044), while the declines per such position did not rise (3.22 → 3.03 and 4.51 → 3.41; three is one for each of Monet’s seats; Table 6). The vector moved the exposure, not the counter. The other two are SESTINA’s: `opPoll` at one seed by 0.05%, where the count is three polls per recorded event on every cell (2.9996 to 2.9999) and that seed played the most events; and `opForced` at another, the seed with the most forced endgames of the twelve.
- *One cell of 72 had no calibration file.* The text harness that prints the readout crashed on a cell where one of 36 per-process files was empty. §6.2 words the control as a per-decile readout on at least 20,000 decisions on every cell, and the per-process files carry it: 53,276 decisions on that cell, all ten deciles populated, the same bias as its neighbours. §3.8at scored twelve cells of the v0.33 panel, whose text harness had crashed the same way, as produced; the reader added the file as a requirement on a note in the registration that misread why those twelve files were missing (§11).

The record keeps the reader’s line verbatim and scores condition 5 as registered; the owner may hold v1.0 to the reader’s line instead, and the record says so.

Condition 6. The second adapter was built independently at §3.8m: its view builder is its own, and it shares only the policy it wraps. It played SESTINA on the same twelve seeds from the same tree. Its win rate equals the arm’s to four decimals on every seed (58.3819% pooled on both), every other engine line is equal on every cell, and all 14,400 games are byte-identical when matched on deal and rotation; the engine’s own count of Monet’s asks is identical on 12 of 12 cells. At v0.33’s

Table 6: Where the `mustfixDeclines` band misses come from. *Dead positions*: positions before the clinch at which FishAI’s rule for a dead game holds on the public log (no hit for 12 events and no claim for 24, or no claim for 60), counted from the engine’s records. *Declines*: the adapter’s count before the clinch, per complete cell.

opponent	vector	dead positions a game	declines a cell	declines per dead position
v0.2	v0.33	0.134	517	3.22
	v0.54	0.346	1,261	3.03
v0.3	v0.33	0.044	231	4.51
	v0.54	0.118	489	3.41
v0.4	v0.33	5.154	16,455	3.01
	v0.54	5.366	19,365	3.01
SESTINA	v0.33	0.036	212	4.68
	v0.54	0.062	316	4.23

vector the twin had differed on 4 games of 14,400, all at one position that changed no result (§3.8at); here no game differs.

The predictions. Each condition carried a probability before the read: 97, 95, 65, 93, 85 and 85%, and about 45% for all six together, most of the doubt on condition 3. All six held, so each prediction hit, condition 5 by the registration’s rules rather than by its reader’s line. One read cannot say whether 45% was well calibrated. It does say that the doubt the registration placed on condition 3 did not materialise: SESTINA sat 0.15 points above v0.6, not more than the floor.

What was seen early. Before the readers ran, the engine lines of four of the 84 cells had been looked at: two v0.3 cells, in the lane log around the calibration crash, and one SESTINA cell and one twin cell while this record was being prepared. The readers had been fixed by md5 before any of them, and nothing in the scoring could move.

9 Monet beside Bass, SESTINA and Kraken

Table 7 sets Monet v1.0 beside the three agents it is most often compared with. Bass v2.0 is FishAI’s previous line [6]: nine hand-tuned play styles and an adaptive engine that classifies each opponent seat from the public log and best-responds. SESTINA v1.0 is described in §2. Kraken v1.0 is the later configuration of KV’s FishBot v0.6, a Python agent that FishLab’s engine runs over the same JSON protocol; its specification, as FishLab’s engine source records it, is FishBot v0.6’s six parameters (`n_draws` 480 sampled deals, `opponent_gamma` 0.35, a lookahead setting of 0.25 with depth 3 and beam 4, `endgame_m` 0) plus `claim_forced_exhaustive`, which enumerates the declaration when one is forced. We have not played Kraken, and its column is a description, not a measurement.

What each design buys. SESTINA spends its compute where the belief is sharpest, in the endgame, and plays a tuned linear policy elsewhere; Kraken searches every decision. Monet spends its compute before the game: 1.16 million imitated decisions and 3.0 million paired play-outs, compiled into two networks of about 7,500 weights each. The record’s own search results (§6) are the argument for that choice against SESTINA: every play-time search Monet tried was limited by its model of the other players and by its evaluation of positions, and the learned correction was not.

Table 7: Four agents, by what decides their asks and declarations. Monet’s row is v1.0.

	Bass v2.0	SESTINA v1.0	Kraken v1.0	Monet v1.0
idea	adapt to the opponent	a tuned formula, and search at the end	sample deals and look ahead	learn from SESTINA, then from outcomes
the ask	one of nine hand-tuned styles, chosen per opponent seat	linear score over 20 features, weights by cross-entropy search; one-step lookahead	Monte Carlo over 480 sampled deals; beam lookahead, depth 3	a clone of SESTINA’s choice, overruled by a model of paired outcomes
search at play	none	endgame only (≤ 26 cards unresolved): 12 deals, top 4 asks, 12 moves	every decision	none
declaring	style thresholds	team ownership ≥ 0.849 and a value check	exhaustive when forced	certain first; $p \geq 0.775$; the planner’s best when compelled
vs SESTINA	27.08%	—	not measured	58.38% (§8)
cost	fast	not measured here	not measured	0.178 ms a decision

Where Monet is weaker. Monet does not calculate: in a forced endgame it follows its policy where a search would enumerate. Half of its ask policy is a copy, and the copy is only as good as SESTINA. The correction was trained on self-play with v0.33 at every seat, so an opponent unlike anything in that population could expose it; the panel read of §8 is the guard against that, and it covers five frozen FishLab releases, not people.

10 Limitations

One opponent family. Every abroad number is against FishLab’s bots in FishLab’s engine; nothing here measures play against people or against Kraken. *One rule set.* **us54** as translated into a host that plays nine half-suits where **us54** clinches at five; the two dialect translations (PASSFIX and MUSTFIX) are measured and disclosed, and numbers from different bridges are never subtracted. *A second adapter, not a second engine.* Condition 6 reproduces the bridge; the engine is FishLab’s on both sides. *The correction’s labels.* adv-4 learned from play-outs of v0.33 in all six seats, so the gain at home is measured against the policy its labels came from; why it carries to SESTINA is not established by any read here. *Statistical resolution.* Twelve seeds resolve ± 2.00 paired; the acceptance read is one read, and its conditions were scored once. *The ancestors are frozen.* Beating v0.2–v0.6 is evidence against having fitted SESTINA alone, not evidence of strength.

11 Corrections to our own record

Writing this paper and reading the acceptance test found errors in MONET.md. The first two are corrected where they stand, with the old text kept beside the correction; the last two are recorded in the acceptance read’s record (§3.8ba), beside the note they correct.

1. +3.64 **cited for** +7.24. Thirteen places in §3.8an–§3.8as cite “§3.8ac’s +3.64 at the bridge” for the greedy ranker’s loss to the clone. +3.64 is §3.8af’s read of v0.33 over v0.30; the clone

over the vector before it read +7.24 (§3.8ac). Every sentence argued from the direction, which stands, and no bar or decision was computed from the size.

2. **SESTINA’s spec keys read as its belief.** Six passages (§3.8b, §3.8f, §3.8af, §3.8ag, §3.8ah, §3.8ak) read `det`, `cand`, `depth`, `kappa`, `maxq` and `rbelief` as properties of SESTINA’s own belief. They configure its endgame search (§2); what the record measured under those names was FishAI’s own mechanism and stands as such.
3. **The calibration harness at the v0.33 panel read, unrecorded.** §3.8at scored the calibration readout as produced on 72 of 72 cells, from the per-process files, which is where the readout’s numbers come from. Neither it nor §3.8ar recorded that the text harness had crashed on 12 of those cells, each time on a per-process file that did not parse when it ran, so that 12 text readouts were never written. The same crash recurred once in this paper’s read (§8).
4. **A pre-registration note.** The note recording the acceptance read’s readers said the condition-5 reader “reads `-version`-free cover files only”; it reads every per-process file in each cell’s cover directory and has no version filter. It also explained the reader’s 60 of 72 on the v0.33 cells by twelve text files “not on disk”, without the crash behind them, so the reader’s calibration row did not reproduce §3.8at’s. Both sentences are corrected in MONET.md beside the registered text.

Data and code

The engine, the policy and both networks (as data modules), the probes, the tests that pin every shipped vector and its forward bank, and MONET.md with every cell of the ladder are in the FishAI repository. The bridge — FishLab’s engine built from source, the recording fork, the two adapters and the game records — is outside it, because FishLab’s repository carries no licence file; it is archived by the owner outside the public tree, and every number here can be re-read from it.

References

- [1] FishAI project. MONET.md — the roadmap from Monet v0.1 to v1.0. Repository document, 2026. §0 the target; §3.1–§3.4 the belief rungs; §3.8a–§3.8az the ladder; §3.8ba the acceptance read; §3.9 the definition of v1.0; §6.2 the controls; §8.3 the decisions.
- [2] A. Hsieh. Toward the frontier: seventeen measured rungs against a frontier Canadian Fish agent, and the wall where they stopped. FishAI project, 2026. The first seventeen rungs and the v0.9 acceptance read.
- [3] A. Hsieh. Against the frontier: a cross-engine measurement and the bridge defect that nearly buried it. FishAI project, 2026. The 27.08% baseline, the mirror-cell lesson and the corrected bridge.
- [4] dylann4500. FishLab — a Canadian Fish engine, bot family and evaluation lab. GitHub repository, <https://github.com/dylann4500/FishLab>. The C++ engine, the SESTINA v1.0 frozen specification, the v0.2–v0.6 lineage, the guest-bot protocol, the Kraken integration, and the attribution of SESTINA’s scoring terms. The repository carries no licence file; nothing from it is reproduced here, and its measured findings are cited as its own.
- [5] FishAI project. RULES_US54.md — the “US student” 54-card variant. Repository rules specification, 2026.
- [6] A. Hsieh. FishAI v2.0: the three-sided ask. FishAI project, 2026. The Bass v2.0 policy Monet v0.1 forked.

A The ladder, rung by rung

Every numbered entry of MONET.md’s ladder, 2026-09-01 to 2026-09-18. *Shipped* means in the registry. Points are paired against the named base, abroad, on twelve fresh seeds unless marked.

rung	idea	outcome
v0.1	Bass v2.0 under the Monet name; the instrument	shipped: 27.08% (§3.1)
v0.2	dead asks set aside; two scorer fixes	shipped: 27.21%, “must not move” (§3.2)
v0.3	licence conditioning at $\lambda = 0.6$; the defusal appetite frozen; score-conditioned urgency measured	shipped: 30.96%, +3.75 on six seeds (§3.3)
v0.4a	the fitted card-by-seat marginal	shipped: 31.94%, +0.99 unresolved (§3.4a)
v0.4b	the joint chain for declares	shipped: 32.75%, +0.81 (§3.4b)
v0.4c	the licence back at $\lambda = 0.3$	shipped: 35.12% on 24 seeds (§3.4c)
—	the capability readout; the gate; the lab arm	read; “the third row”; not built (§3.5)
v0.5	opponent reading	+0.47 (SE 0.59) (§3.6)
v0.6	communication: revealing asks, the handoff	flat or behind at home (§3.7)
v0.7	the search arm	+0.14 (SE 0.67) (§3.8a)
v0.8	the determinized declare	+0.08 (SE 0.08) (§3.8b)
—	the attribution study	read (§3.8c)
v0.9	the contest credit	shipped: 38.92%, +4.04 (§3.8d)
v0.10	the exposure charge	+1.07 (SE 0.54); re-read −0.15 (§3.8e)
v0.11	the declare priced; MUSTFIX adopted	+1.28 (SE 0.08), a bridge correction (§3.8f)
—	the majority conversion	records study (§3.8g)
v0.12	the closing ask	+0.58 (SE 0.23), under the floor (§3.8h)
v0.13	the chase appetite	rejected on the fit (§3.8i)
v0.14	the assignment	records study: the axis closes (§3.8j)
v0.15	the deduction fix	records study: dead (§3.8k)
v0.16	the calibrated licence likelihood	−0.74 (SE 0.46) (§3.8l)
v0.17	§3.9 at v0.9’s vector	conditions 1 and 3 not met (§3.8m)
—	publication and the ship rule	the owner’s rule adopted (§3.8n)
v0.18	the stack read: exposure and closing	nothing clears the rule (§3.8o)
v0.19	the even-set race, priced	read (§3.8p)
v0.20	the four-of-six decision	read: $\Delta_{\text{lead}} = +19.6$ (§3.8q)
v0.20b	the race-pace term at home	stopped at home (§3.8r)
v0.20c	the closing credits abroad	shipped: 41.25%, +0.94 (§3.8s)
v0.21	the rules-certain count	records read (§3.8t)

rung	idea	outcome
v0.22	the closing credit at three of six	rejected at home (§3.8u)
v0.23	the appetite doses moved one at a time	rejected at home (§3.8v)
v0.24	the placement value	records read (§3.8w)
v0.25	the trailer’s ask	records read (§3.8x)
v0.26	the five rung’s dose above one	rejected at home (§3.8y)
v0.27	the doses moved two at a time	rejected at home (§3.8z)
v0.28	the search at any compute	−1.61 (SE 0.77) (§3.8aa)
v0.29	a learned leaf	never read abroad (§3.8ab)
v0.30	the SESTINA clone	shipped: 48.01%, +7.24 (§3.8ac)
v0.31	the clone refitted wider and longer	stopped at home (§3.8ad)
v0.32	a declare clone	not built (§3.8ae)
v0.33	the clone’s second feature set	shipped: 51.04%, +3.64 (§3.8af)
v0.34	the wider net at the second width	stopped at its fit gate (§3.8ag)
v0.35	a holder model of the deal	stopped at its fit gates (§3.8ah)
v0.36	the search with the clone as its policy	−1.26 points at home (§3.8ai)
v0.37	the decisions the clone does not cover	records replay (§3.8aj)
v0.38	the holder model in the claim path	+0.28 ± 0.50 at 2 SE (§3.8ak)
v0.39	where the sets go, at v0.33	read (§3.8al)
v0.40	SESTINA’s gambles	read (§3.8am)
v0.41	the lead conversion under the clone	read: the gap closed (§3.8an)
v0.42	the belief’s ceiling at our decisions	read (§3.8ao)
v0.43	the repair’s reach; the ceiling above the clone	read (§3.8ap)
v0.44	the search over the clone’s shortlist	not built: stopped at its gate (§3.8aq)
v0.45	§3.9 condition 3 at v0.33	met (§3.8ar)
v0.46	a value fitted on winning, over the shortlist	rejected at home (§3.8as)
v0.47	§3.9 conditions 5 and 6 at v0.33	met (§3.8at)
v0.49	the learned leaf inside the search	rejected at its marker (§3.8au)
v0.50	the two ceilings	no belief-limited gain clears zero (§3.8av)
v0.51	the learned advantage, pilot (adv-1)	its marker missed (§3.8aw)
v0.52	the second round (adv-2)	held at home (§3.8ax)
v0.53	adv-2 at margin 0.2	shipped: 52.05%, +1.46 (§3.8ay)
v0.54	adv-4 at margin 0.05	shipped: 58.42%, +6.25 at 800 deals (§3.8az)
v0.55	§3.9 at v0.54’s vector	all six met, condition 5 as registered: v1.0 is v0.54’s vector (§3.8ba)